গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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SUB-SOIL INVESTIGATION REPORT FOR PROPOSED CONSTRUCTION OF 05 STORIED RESIDENTIAL BUILDING AT TEMUKHI, KUMARGAON, SYLHET DAG # 374, KHA # 9623, J.L # 80, MOUJA- KUMARGOAN

CLIENT

MD. SALAH UDDIN
S/O- MD. SHAHAB UDDIN

Verification Code: qa9P0HpG
Verify report at: crtc-cee-sust.com

CONDUCTED BY
CENTER FOR RESEARCH, TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET.

Sample Collected by
M/S TECHNO SOIL INVESTIGATOR
Mokothchhor Market, Bagbari Barnomala Point, Sylhet.
Mobile: 01712-710546, 01612710546

Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science and Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-50 ft	10
02	Disturbed sample	02	0-45 ft	09
03	Disturbed sample	03	0-45 ft	09
04	Disturbed sample	03	0-45 ft	09
05	-	-	-	-
06	-	-	-	-
07	-	-	-	-
08	-	-	-	-
09	-	-	-	-
10	-	-	-	-

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
	5.0-10	4	0.436	0.459	0.500	0.515	0.556	0.605	0.470
BH02	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	4	0.516	0.543	0.592	0.609	0.658	0.716	0.556
BH03	2.5-5.0	4	0.464	0.488	0.532	0.548	0.592	0.644	0.500
	5.0-10	8	0.912	0.959	1.046	1.077	1.164	1.266	0.983
BH04	2.5-5.0	4	0.487	0.512	0.559	0.575	0.622	0.676	0.525
	5.0-10	9	0.992	1.043	1.137	1.172	1.266	1.377	1.069
BH05	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH06	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH07	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.096	0.096	0.096	0.118	-	-	-	-	-	-
10	0.025	0.025	0.049	0.049	-	-	-	-	-	-
15	0.049	0.074	0.138	0.138	-	-	-	-	-	-
20	0.174	0.196	0.246	0.240	-	-	-	-	-	-
25	0.128	0.136	0.240	0.118	-	-	-	-	-	-
30	0.160	0.232	0.168	0.232	-	-	-	-	-	-
35	0.216	0.224	0.176	0.224	-	-	-	-	-	-
40	0.216	0.216	0.216	0.216	-	-	-	-	-	-
45	0.208	0.208	0.208	0.208	-	-	-	-	-	-
50	0.200	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.900	0.900	0.900	1.125	-	-	-	-	-	-
10	0.225	0.225	0.450	0.450	-	-	-	-	-	-
15	0.450	0.675	1.350	1.350	-	-	-	-	-	-
20	1.800	2.025	3.375	3.825	-	-	-	-	-	-
25	25.600	27.200	3.825	1.125	-	-	-	-	-	-
30	32.000	46.400	33.600	46.400	-	-	-	-	-	-
35	43.200	44.800	35.200	44.800	-	-	-	-	-	-
40	43.200	43.200	43.200	43.200	-	-	-	-	-	-
45	41.600	41.600	41.600	41.600	-	-	-	-	-	-
50	40.000	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.062	0.062	0.062	0.077	-	-	-	-	-	-
10	0.016	0.016	0.032	0.032	-	-	-	-	-	-
15	0.032	0.048	0.090	0.090	-	-	-	-	-	-
20	0.113	0.127	0.160	0.156	-	-	-	-	-	-
25	0.083	0.088	0.156	0.077	-	-	-	-	-	-
30	0.104	0.151	0.109	0.151	-	-	-	-	-	-
35	0.140	0.146	0.114	0.146	-	-	-	-	-	-
40	0.140	0.140	0.140	0.140	-	-	-	-	-	-
45	0.135	0.135	0.135	0.135	-	-	-	-	-	-
50	0.130	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.405	0.405	0.405	0.506	-	-	-	-	-	-
10	0.101	0.101	0.203	0.203	-	-	-	-	-	-
15	0.203	0.304	0.608	0.608	-	-	-	-	-	-
20	0.810	0.911	1.519	1.721	-	-	-	-	-	-
25	11.520	12.240	1.721	0.506	-	-	-	-	-	-
30	14.400	20.880	15.120	20.880	-	-	-	-	-	-
35	19.440	20.160	15.840	20.160	-	-	-	-	-	-
40	19.440	19.440	19.440	19.440	-	-	-	-	-	-
45	18.720	18.720	18.720	18.720	-	-	-	-	-	-
50	18.000	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed five storied residential building.

Soil profile scenario

Overall soil information of the investigated site is quite irregular in between the boreholes and also vertical inconsistency exists in the soil formation. In fact, clay soil with low bearing capacity up to top 20 ft has been observed in most of the boreholes whereas sandy soil with high bearing capacity from 30 ft onward has been found in all boreholes.

Foundation options

The soft soil from the top 10 ft depth is to be excavated first and then a compacted bed of a minimum 5 ft thickness should be prepared using brick/stone chips and sand. Finally, the foundation and superstructure construction should be carried on following the intermittent construction process. However, in this case, the building plinth level should be kept quite higher and settlement must be monitored in regular basis. Moreover, deep foundation (Bored/precast pile) may be selected to ensure the stability of the structure and to overcome the ground improvement complication. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found from the attached table (8 and 9) in the report which is based on the existing soil condition.

Safety provisions

The earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. Client performed no special laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **45-50** ft as client provided.

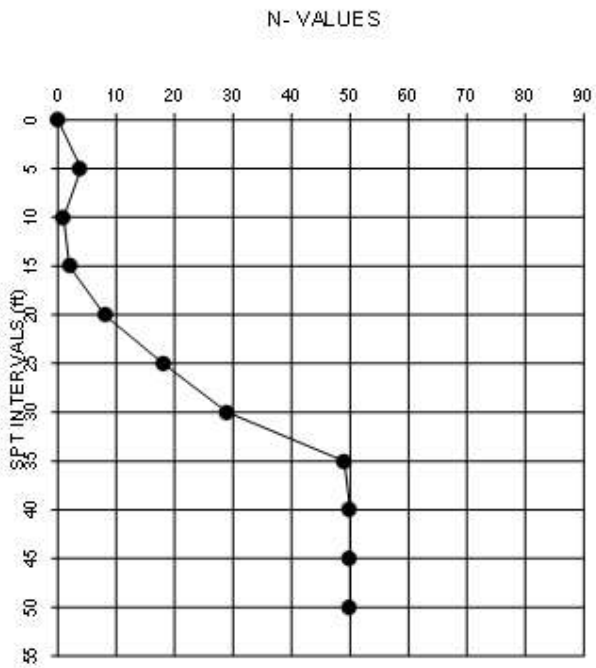
Test Assisted By	Report Prepared By	Countersigned By
A. H. M. Humayun Kabir	**Dr. Md. Imran Kabir**	**Prof. Dr. HMA Mahzuz**
Senior Technical Officer Dept. of CEE, SUST	Professor Dept. of CEE, SUST	Coordinator, CRTC Dept. of CEE, SUST

RECORD OF BORING AND TESTING

CLIENT : **MD. SALAH UDDIN**	LOCATION: Temukhi, Kumargaon, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 05 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 3'-0" BELOW EGL	
BORE HOLE NO. **01**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 27/08/2022	
BORING DIA : **0.33ft**		Sand with silt & clay.
DEPTH OF BORING : 50.0 ft	COMPLETION DATE : 27/08/2022	
		Organic/Gravel

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 Inch	06 Inch	06 Inch	N- Value
						0	0	0	0	0
D1		5.0	20.0	Grey soft clay		5.0	1	2	2	4
D2		10.0				10.0	1	0	1	1
D3		15.0				15.0	1	1	1	2
D4		20.0				20.0	3	4	4	8
D5		25.0	30.0	Grey fine sand with silt		25.0	6	8	10	18
D6		30.0				30.0	7	15	14	29
D7		35.0				35.0	14	21	28	49
D8		40.0				40.0	22	40	10	50
D9		45.0				45.0	19	25	25	50
D10		50.0				50.0	18	50	over	50

GRAPHICAL REPRESANTATION OF N- VALUES

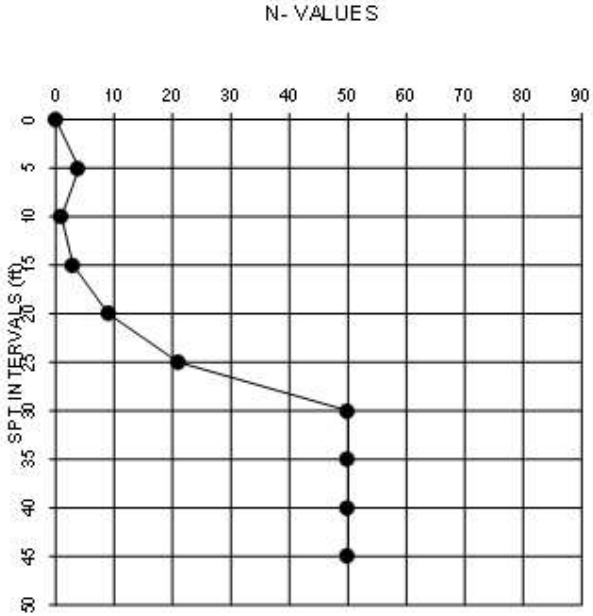


RECORD OF BORING AND TESTING

CLIENT : **MD. SALAH UDDIN**	LOCATION: Temukhi, Kumargaon, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 05 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 3'-0" BELOW EGL	
BORE HOLE NO. **02**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 27/08/2022	
BORING DIA : **0.33ft**		Sand with silt & clay.
DEPTH OF BORING : 45.0 ft	COMPLETION DATE : 27/08/2022	
		Organic/Gravel

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 Inch	06 Inch	06 Inch	N- Value
						0	0	0	0	0
D1		5.0	20.0	Grey soft clay		5.0	1	2	2	4
D2		10.0				10.0	1	0	1	1
D3		15.0				15.0	1	1	2	3
D4		20.0				20.0	2	4	5	9
D5		25.0	25.0	Grey fine sand with silt		25.0	5	9	12	21
D6		30.0				30.0	19	40	10	50
D7		35.0				35.0	14	26	24	50
D8		40.0				40.0	15	30	20	50
D9		45.0				45.0	25	50	over	50

GRAPHICAL REPRESANTATION OF N- VALUES

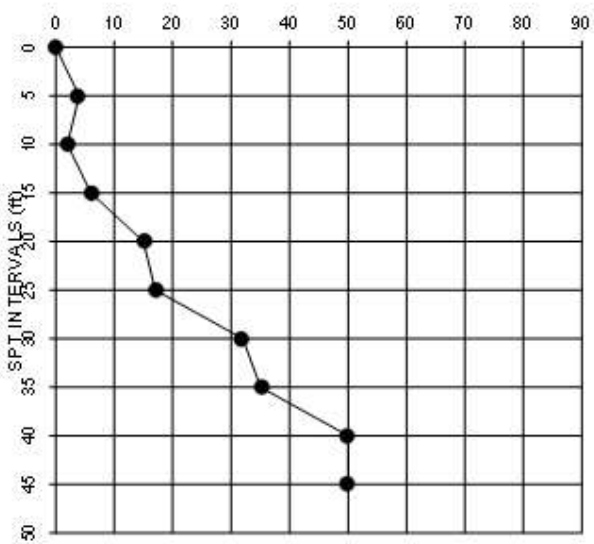


RECORD OF BORING AND TESTING

CLIENT : MD. SALAH UDDIN	LOCATION: Temukhi, Kumargaon, Sylhet.		Sandy clay with silt.
PROJECT NAME : CONSTRUCTION OF 05 STORIED RESIDENTIAL BUILDING.	GROUND WATER LEVEL : 3'-0" BELOW EGL		
BORE HOLE NO. 03			Clay with silt & sand.
METHOD OF BORING : WASH.	STARTING DATE : 27/08/2022		
BORING DIA : 0.33ft			Sand with silt & clay.
DEPTH OF BORING : 45.0 ft	COMPLETION DATE : 27/08/2022		
			Organic/Gravel

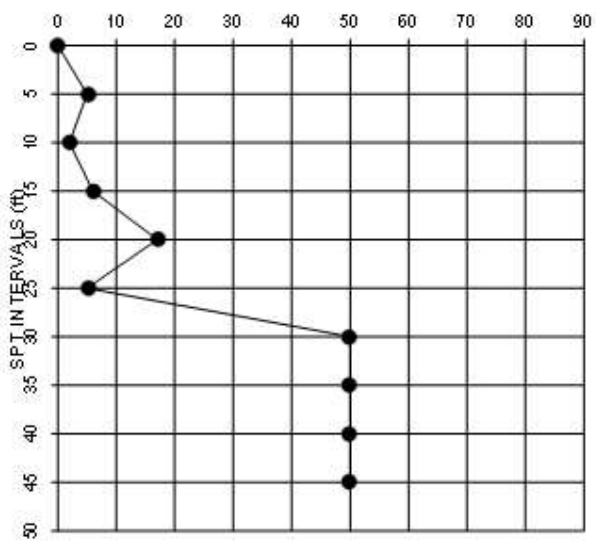
Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 Inch	06 Inch	06 Inch	N- Value
						0	0	0	0	0
D1		5.0	15.0	Grey soft clay		5.0	2	2	2	4
D2		10.0				10.0	1	1	1	2
D3		15.0				15.0	1	3	3	6
D4		20.0	10.0	Grey silty clay		20.0	4	6	9	15
D5		25.0				25.0	6	9	8	17
D6		30.0	20.0	Grey fine sand with silt		30.0	9	20	12	32
D7		35.0				35.0	8	17	18	35
D8		40.0				40.0	15	32	18	50
D9		45.0				45.0	21	50	over	50

GRAPHICAL REPRESANTATION OF N- VALUES



RECORD OF BORING AND TESTING

CLIENT : **MD. SALAH UDDIN**	LOCATION: Temukhi, Kumargaon, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 05 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 3'-0" BELOW EGL	
BORE HOLE NO. **04**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 27/08/2022	
BORING DIA : **0.33ft**		Sand with silt & clay.
DEPTH OF BORING : 45.0 ft	COMPLETION DATE : 27/08/2022	
		Organic/Gravel

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 Inch	06 Inch	06 Inch	N- Value
						0	0	0	0	0
D1		5.0	15.0	Grey soft clay		5.0	2	2	3	5
D2		10.0				10.0	1	1	1	2
D3		15.0				15.0	2	3	3	6
D4		20.0	5.0	Grey silty clay		20.0	5	7	10	17
D5		25.0	5.0	Grey soft clay		25.0	2	2	3	5
D6		30.0	20.0	Grey fine sand with silt		30.0	11	39	11	50
D7		35.0				35.0	13	40	10	50
D8		40.0				40.0	15	50	over	50
D9		45.0				45.0	30	50	over	50

GRAPHICAL REPRESANTATION OF N- VALUES